

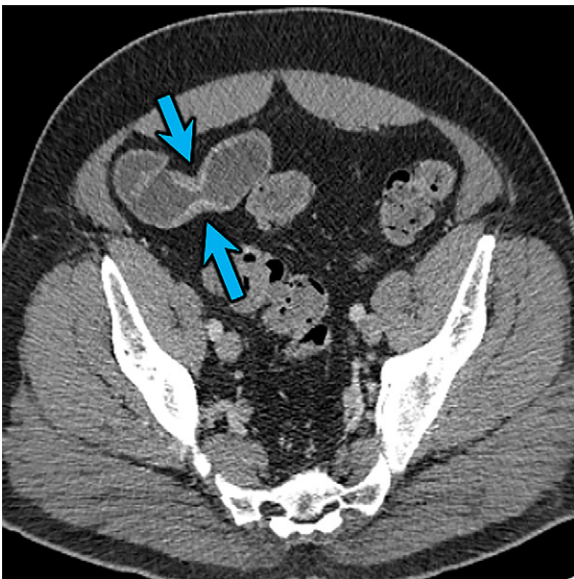
# Fundamentals of Small Bowel Imaging: What Radiology Residents Should Know

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Author affiliations, funding, and conflicts of interest are listed at the [end of this article](#).  
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**Figure 1.** Small bowel lymphoma in a 42-year-old man with fatigue, night sweats, and anemia. Axial contrast-enhanced CT image shows aneurysmal dilatation (arrow) of a markedly thickened loop of small bowel (arrowheads) in the central abdomen. Biopsy results confirmed diffuse large B-cell lymphoma.



**Figure 2.** NSAID enteropathy in a 60-year-old man receiving daily meloxicam therapy for fibromyalgia. Axial contrast-enhanced CT enterographic image shows a short-segment stricture of the ileum (arrows).

Cross-sectional and fluoroscopic imaging play a pivotal role in evaluating the jejunum and ileum because of less accessibility for endoscopic evaluation compared with the stomach, duodenum, or colon. For the radiologist in training, assessment of the small bowel can be challenging, in large part due to the overlapping imaging features of many small bowel diseases. A pattern-based approach to small bowel imaging can be useful in potentially narrowing the differential diagnosis and allowing the radiologist to offer a more useful impression.

The slide presentation comprises an extensive and illustrative review of clinical and radiologic aspects of multiple small bowel diseases. These are classified, for didactic purposes, into seven main categories: (a) benign diseases: diverticulitis, celiac sprue, Crohn disease, scleroderma, and infection; (b) trauma and hemorrhage: blunt trauma, penetrating trauma, luminal hemorrhage, and intramural hemorrhage; (c) vascular disease: ischemia and vasculitis; (d) obstruction: mechanical and closed-loop obstruction; (e) treatment related: graft versus host disease, nonsteroidal anti-inflammatory drug (NSAID) enteropathy, angioedema, radiation enteritis, and lymphangiectasia; (f) benign neoplasms: polyposis, gastrointestinal stromal tumor (GIST), and lipoma; and (g) malignant neoplasms: adenocarcinoma, GIST, carcinoid, lymphoma, and metastatic disease.

Although cross-sectional imaging is emphasized in the slide presentation, fluoroscopic examples are presented when available. Features shared across disease entities, such as aneurysmal luminal dilatation of masses exhibiting an exoenteric growth pattern (Fig 1) and features specific to a disease, such as multiple short-segment strictures or diaphragms in NSAID enteropathy (Fig 2), are shown.

The radiologist should be aware of the spectrum and typical imaging appearances of small bowel diseases to accurately guide clinical management.



RadioGraphics 2023; 43(2):e220094  
<https://doi.org/10.1148/rg.220094>

Content Codes: FL, GI

## TEACHING POINTS

- The jejunum and ileum are less accessible than the stomach, duodenum, and colon for endoscopic evaluation. Cross-sectional and fluoroscopic imaging play a key role in detection and diagnosis of small bowel disease.
- A pattern-based approach integrating the appearance of wall thickening, degree and symmetry of thickening, length and segment of involvement, and clinical history can hone an otherwise broad differential diagnosis into a more useful impression.
- Awareness of the spectrum and typical appearance of small bowel diseases facilitates the radiologist guiding clinical management.

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**Disclosures of conflicts of interest.**—D.H.B. Editorial board member of *Radiographics*. All other authors have disclosed no relevant relationships.

## Suggested Readings

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